Check for updates

Received: 30 March 2020 | Revised: 24 December 2020 | Accepted: 16 January 2021

DOI: 10.1111/coa.13729

ORIGINAL ARTICLE

WILEY

Long-term therapy with botulinum toxin in facial synkinesis: Retrospective data analysis of data from 1998 to 2018

Sara Alipour[1] | Carina Pick[1] | Stefanie Jansen[1] | Svenja Rink[3] | Jens Peter Klußmann[1,2] | Maria Grosheva[1]

[1]Department of Otorhinolaryngology, Head and Neck Surgery, Medical Faculty, University of Cologne, Cologne, Germany

[2]Center for Molecular Medicine Cologne (CMMC), University of Cologne, Faculty of Medicine and University Hospital Cologne, Cologne, Germany

[3]Department of Prosthetic Dentistry, School of Dental and Oral Medicine, University Hospital, University of Cologne, Cologne, Germany

Correspondence
Carina Pick, Department of Otorhinolaryngology, Head and Neck Surgery, Medical Faculty, University of Cologne, Kerpener Str. 62, 50937 Cologne, Germany.
Email carina.pick@uk-koeln.de

Abstract

Objectives: Treatment with botulinum toxin A (BoNT) is the therapy of choice for many patients with facial synkinesis. Repeated injections relieve hypertonicity and hyperkinesis of reinnervated mimic muscles. Aim of the study was to prove if the injection regime and dosage of BoNT change during long-time therapy.

Design: Retrospective analysis of patients´ data, who were treated for synkinesis with BoNT from 1998 to 2018.

Setting: Tertiary referral facial nerve centre.

Participants: Injection pattern of BoNT was based on clinical symptoms, observations of the specialist and on previous treatment pattern. Onabotulinumtoxin (OnaBoNT), Incobotulinumtoxin (IncoBoNT) and Abobotulinumtoxin (AboBoNT) were available for treatment. Patients consulted our department for following treatment as soon as the symptoms re-occurred.

Main outcome measures: Change in dosage and injection pattern, the time intervals between treatments over the entire therapy period.

Results: Seventy-three patients were repeatedly injected. The median number of treatments was 18, the median treatment interval was 3.0 months. During the initial treatment, orbicularis oculi and the mentalis muscles were the most frequently injected muscles (94%). During repeated treatment, the number of injected muscles increased significantly ($P < .0001$), whereas the dose per muscle remained stable. The initial dose was 24 U (95%-CI 22-27 U) for OnaBoNT and IncoBoNT; 69 U for AboBoNT(95%-CI 44-94 U). We observed a significant increase in dosage for OnaBoNT and IncoBoNT ($P < .0001$) during the long-term therapy. The time intervals between treatments were stable for all three BoNT preparations ($P > .05$).

Conclusions: We observed significant change in treatment dose and injection pattern of BoNT in patients with facial synkinesis. These results provide an orientation in dose finding and injection regimen of BoNT in the long-term course of therapy.

Sara Alipour and Carina Pick contributed equally to the manuscript and share the first authorship.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.
© 2021 The Authors. Clinical Otolaryngology published by John Wiley & Sons Ltd.

1 | INTRODUCTION

Treatment with BoNT is an established and effective therapy for patients with facial synkinesis worldwide.[1-5] Despite of its proven efficacy, the BoNT therapy is still declared an off-label use. In several countries, patients have to bear the costs for the repeated injections themselves, because they are not covered by the health insurance companies.[6,7]

The injection pattern in a patient with synkinesis is often based on the recommended injection treatment for mimic wrinkles or for blepharospasm.[8-10] However, the leading symptom in facial synkinesis is not a facial asymmetry. For most patients, the treatment of involuntary movements, hyperactivity, twitching, spasticity of the muscles and hypertonicity is more important than achievement of perfect facial symmetry.[4,11]

Several working groups previously reported about wide-ranging effects of BoNT in facial synkinesis.[4,12,13] However, only few information exists on the injection patterns and the effects of the BoNT in the long-term therapy. Aim of the current study was to evaluate the injection pattern and evolution in dosage of BoNT during long-term treatment.

Keypoints

- Treatment with botulinum toxin A is the therapy of choice for patients with facial synkinesis despite its off-label application.
- The injection pattern and treatment frequency are usually based on the recommendations from cosmetic applications and from the experience in treatment of blepharospasm.
- The present data showed a shorter duration of effect of BoNT in facial synkinesis compared with its cosmetic application. These results underline the secondary role of esthetic symmetry in treatment of synkinesis.
- The presented results provide an orientation in dose finding and injection regimen of BoNT in patients with facial synkinesis during a long-term therapy.

2 | MATERIAL AND METHODS

2.1 | Data collection

The data of all patients, who were treated in the tertiary referral facial nerve centre with facial synkinesis from 1998 to 2018 were retrospectively evaluated. Three brands of BoNT were available for therapy: Abobotulinumtoxin A, AboBoNT (Dysport®; Ipsen, Germany) from 1998 till 12/2004; Onabotulinumtoxin A, OnaBoNT(Botox®, Allergan, Ireland) since 03/2004; and Incobotulinumtoxin A, IncoBoNT (Xeomin®, Merz, Germany) since 2008. Before use, 100 units of OnaBoNT or IncoBoNT were reconstructed with 2 mL sterile 0.9% sodium chloride (NaCl) solution; 500 units of AboBoNT were reconstructed in 2.5 mL NaCl. There with, 0.1 mL injection solution corresponded to 5 units of OnaBoNT or IncoBoNT; and 0.1 mL injection solution of AboBoNT corresponded to 20 units.

All patients were informed about the possible side effects and about the off-label use of the BoNT before the first treatment. The pre-treatment face was photodocumented and discussed with the patient. Considering the patient's current complaints (what is your most disturbing symptom? which result do you wish to achieve? etc) an injection plan was developed and documented. To obtain a satisfactory result, treatment of the upper and the lower face muscles on the synkinetic side was recommended. The basic therapy plan included the injection of orbicularis oculi, mentalis and platysma muscles. In severe ocular synkinesis, the injection of procerus and/or corrugator supercilii and the frontal muscles were recommended. In cases of increased hypertonicity of the middle face, injection of buccinator muscle was carried out. To optimise the result in the midface, nasalis, zygomaticus major and levator labii superioris muscles were optionally treated with low dosages of BoNT. To achieve facial symmetry, injection of contralateral procerus and/or corrugator, mentalis and depressor anguli oris muscles was carried out. However, some patients still preferred the treatment of the synkinetic side only. The injection pattern was adapted during every visit to current symptoms of the patient, to the degree of synkinesis and to side effects or unfavourable outcome from previous injections such as asymmetry or muscle dysfunction.

The injections were carried out without electromyographic guidance. For each patient, the date of the visit, number of injected muscles, respective dose per muscle and specific injection points as well as the total dose and the used formulation of BoNT were documented on a centre-specific BoNT injection record. Patients consulted our department for the next treatment, as soon as they felt an increasing muscle tension of the synkinetic muscles, abnormal spasms, twitching or increasing asymmetry of the face. Experienced patients in a long-term therapy sometimes made an appointment in advance. Several different ENT-specialists which were specialised in BoNT therapy and in treatment of facial palsy carried out the injections during the analysed time-period. However, care was taken, that one patient was treated by one specialist only to obtain perfect long-term results.

We analysed the change in the total dosage of BoNT during repeated treatments, change of dose per muscle and the number of treated muscles, as well as the time periods between treatments.

2.2 | Statistical evaluation

We used IBM SPSS Software, Version 25 (IBM Inc, Chicago, IL; USA) for the analysis of the data. Unless otherwise noted, the quantitative data is displayed as mean ± standard deviation (SD), the qualitative data is expressed as absolute numbers and percent. Quantitative

variables for non-normally distributed data were calculated using the Mann-Whitney U or Kruskal-Wallis H test. We applied Friedman test for analysis of dependent variables measured during multiple time points on the same subjects (number of treated muscles and change of BoNT dosage during repetitive treatments). Qualitative variables were evaluated using the Fisher´s exact or Pearson´s chi-squared test. A *P*-value of <.05 was considered statistically significant. All *P*-values are two-sided.

Only patients with repeated treatments (73 of 84) were included into analysis of the time intervals between treatments. To analyse the change in dosage and the number of injected muscles, only patients with initial treatment in our department (66 of 84) were evaluated.

3 | RESULTS

3.1 | Patients´ characteristics

The database included 84 patients who were treated 840 times from 1998 to 2018. Of these, 77% were female (n = 65). The mean age was 49 ± 14.9 years (range, 17-77 years). Seventy-three (73) patients were injected repeatedly. In this group, the median number of treatments was 18, the median treatment interval was 3.0 months. Sixty-six patients (66) were treated for facial synkinesis for the first time in our department, six patients were injected with BoNT before elsewhere. In 12 patients, it was unknown whether this was a first or repeated treatment.

In 57% of the patients, synkinesis occurred due to facial palsy on the right, in 43% on the left side. The aetiology of the facial palsy was known in 63 patients (Table 1). An idiopathic palsy was the most frequent aetiology (38%) followed by varicella-zoster virus infection (22%). For the initial therapy, we used IncoBoNT in 30 patients, OnaBoNT in 29 and AboBoNT in 22 patients. In three patients, the initial BoNT formulation was unknown. During the entire course of treatment (n = 840), we used IncoBoNT in 380 patients, OnaBoNT312 patients and AboBoNT in 74 patients. In 74 cases, the data on the toxin brand was missing.

3.2 | Pattern of treatment unrelated to the synkinetic side

3.2.1 | Side of treatment

At the initial treatment, 37% (n = 30) of the patients were treated on the right side only, 22% (n = 18) on the left side only and 41% (n = 33) on both sides (side unknown in three patients). Bilateral injections were carried out more frequently during repetitive therapy. During the entire course of treatments, only right side was injected in 19% (n = 156), only left side in 15% (n = 126) and bilateral injections were carried out in in 66.3% (n = 554). The side was unknown in four patients.

3.2.2 | Number of injected muscles

During repeated BoNT treatments, the number of the injected muscles increased significantly (5.8 ± 3), compared with the initial treatment (initial treatment 4 ± 2.2, $P < .0001$). Table 2 shows the number of treated muscles on the right or left side during the initial treatment in our department (n = 66) and during all treatments (n = 761), independent of the side of synkinesis. The number of injected muscles per treatment increased significantly during the first 16 visits (Figure 1, Friedman test, $P < .0001$).

At the initial treatment, orbicularis oculi and mentalis muscles were most frequently injected (50% and 44% on the right and 41% and 44% on the left side, respectively). Corrugator, procerus muscles and platysma (on right and left side) were treated in over 20% of the cases (Table 2). Comparing this data with the total number of treatments, the injection regime changed during the long-term therapy. Frontalis ($P < .0001$), procerus ($P < .003$), orbicularis oculi ($P < .007$), nasalis ($P < .006$), orbicularis oris ($P < .028$), mentalis ($P = .001$), platysma ($P = .010$) and procerus ($P = .002$) muscles were injected more frequently. Furthermore, the initial pattern was extended by additional target muscles (Table 2).

TABLE 1 Aetiology of the facial nerve paresis in 63 patients

Aetiology of the facial nerve palsy	Percent (%)	Number
Idiopathic facial nerve palsy	38.1	24
Bell's palsy	34.9	22
Mona Lisa Syndrome	3.2	2
Postoperative	31.8	20
Vestibular Schwannoma	9.5	6
Parotidectomy with Facial Nerve reconstruction (malignant lesion)	4.8	3
Superficial/partial parotidectomy (benign lesion)	6.3	4
Total parotidectomy (benign lesion)	1.6	1
Revision-parotidectomy (benign lesion)	1.6	1
Parotidectomy with Facial Nerve reconstruction (benign lesion)	1.6	1
NXII-NVII-transfer	1.6	1
Mastoidectomy	1.6	1
Tumour of the jugular glomus	1.6	1
Revision surgery of the ear fistula	1.6	1
Infectious	23.8	15
VZV	22.2	14
Borreliosis	1.6	1
Posttraumatic	6.4	4
Fracture of the temporal bone	1.6	1
Fracture of the mandibular joint/ midface trauma	3.2	2
Posttraumatic, not otherwise classified	1.6	1
SUM	100	63

TABLE 2 Number of treated muscles on the right and the left face at the initial visit (initial) and during the total course of therapy (total), independent of the side of synkinesis

	Right side					Left side				
	Initial (n = 66)		Overall (n = 761)			Initial (n = 66)		Overall (n = 761)		
Mimic muscles	percent	N	percent	N	*P*-value	percent	N	percent	N	*P*-value
FRONTALIS	11%*	7	32%*	241	**<.0001**	9%*	6	19%*	141	.055
CORR	26%*	17	33%*	253	.214	27%*	18	36%*	273	.161
PROCERUS	27%*	18	47%*	354	**.003**	24%*	16	44%*	333	**.002**
ORB OCULI	50%*	33	67%*	507	**.007**	44%*	29	47%*	360	.599
NASALIS	2%*	1	13%*	99	**.006**	3%*	2	10%*	72	.79
ZYGOM MAJ	18%	12	17%	126	.734	15%	10	13%	102	.691
ORB ORIS	6%*	4	16%*	124	**.028**	2%*	1	8%*	60	.058
MENTALIS	41%*	27	62%*	470	**.001**	44%*	29	47%*	359	.614
DEPRESS ANG ORIS	3%*	2	5%*	36	.527	3%	2	4%	30	.713
PLATYSMA	23%*	15	39%*	294	**.010**	21%	14	19%	146	.689
BUCCINATOR	0%*	0	3%*	24	.143	0%*	0	3%*	20	.183
LEV LABII SUP	2%	1	1%	7	.636	2%	1	1%	4	.320
MASSETER	0%	0	0%	0	1	2%	1	1%	4	.320

Note: Frequencies and percentages refer to the muscle treatment carried out or not. *P*-values were calculated with Mann-Whitney *U* test. Significant changes are highlighted (bold).

*Marks the increasing values.

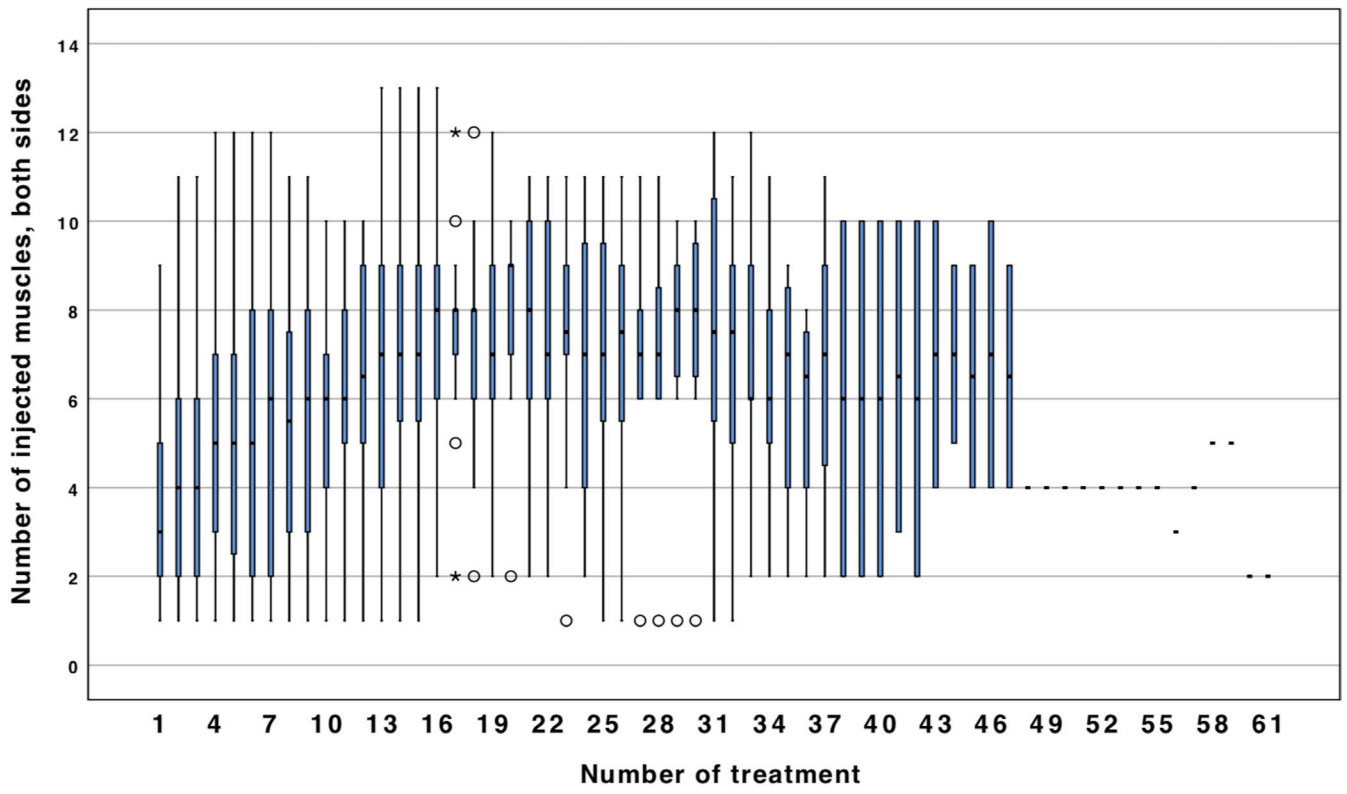


FIGURE 1 Number of injected muscles per treatment. The continuously increasing number of injected muscles is evident during the first 16 visits. Friedman test, $P < .0001$. (Group size N = 1 for treatment number 48-61; group size N < 6 for treatment number 33-47)

TABLE 3 Distribution of treated muscles on the right and the left side in *patients with right-sided synkinesis* during the initial treatment (initial, n = 35) and in the long-term course (overall, n = 506)

	Right side			Left side		
	Initial	Overall	*P*-value	Initial	Overall	*P*-value
FRONTALIS	14.3%	32.9%	**.013**	5.7%	15.4%	.099
CORR	31.4%	34.8%	.622	22.9%	35.4%	.096
PROCERUS	22.9%	51.8%	**<.0001**	14.3%	45.5%	**<.0001**
ORB OCULI	88.6%	95.3%	**.036**	0.0%	15.4%	**.009**
NASALIS	2.9%	19.8%	**.009**	0.0%	0.4%	.697
ZYGOM MAJ	34.3%	27.9%	.364	0.0%	0.4%	.697
ORB ORIS	8.6%	21.5%	**<.0001**	0.0%	0.8%	.582
DEPRESS ANG ORIS	5.7%	4.2%	.594	2.9%	2.4%	.856
LEV LABII SUP	2.9%	1.8%	.448	0.0%	0.0%	1
MENTALIS	57.1%	72.1%	.027	28.6%	34.0%	.447
BUCCINATOR	0.0%	4.7%	.168	0.0%	0.0%	1
PLATYSMA	42.9%	59.5%	**.031**	0.0%	0.6%	.634
MASSETER	0.0%	0.0%	1	0.0%	0.0%	1

Note: P-values were calculated with Mann-Whitney *U* test. Significant changes are highlighted (bold).

3.3 | Pattern of treatment with BoNT in relation to synkinetic side

Distribution of muscles which were treated during the initial visit and during long-term therapy depending on the side of synkinesis is shown in the Tables 3 and 4.

In patients with right-sided synkinesis, orbicularis oculi, mentalis and platysma muscles on the right side were injected in most of patients at the initial visit (88.6%, 57.1% and 42.9%, respectively). Corrugator and mentalis muscles were often injected on both sides. Frontalis, orbicularis oculi, nasalis, orbicularis oris, mentalis muscles and platysma on the right side were treated significantly more often during repeated treatments (all $P < .05$) compared with the initial treatment. On the opposite side (left side), procerus and orbicularis oculi muscles were treated significantly more frequent during long-term therapy ($P < .0001$ and $P = .009$, respectively, Table 3).

In patients with left-sided synkinesis, same muscles (orbicularis oculi, mentalis and platysma) were the most targeted muscles (at initial treatment in 96.7%, 63.3% and 46.7%, respectively). However, only left nasalis and left orbicularis oris muscles were treated significantly more often during repeated treatments (both $P = .028$). On the opposite side (right side), frontalis, orbicularis oculi and mentalis muscles were treated significantly more frequent in the long-term course of therapy (all $P < .05$, Table 4).

3.4 | Change of dosage during the long-term therapy

Mean dosage of OnaBoNT, IncoBoNT and AboBoNT at initial treatment was 26 ± 10.5units, 23.1 ± 9.9 units and 55.4 ± 23.8 units, respectively. Overall dosage for all three formulations was higher than the dosage at the initial treatment (Table 5, significant for IncoBoNT only).

Due to the same equivalent dose of IncoBoNT and OnaBoNT, continuous change in dosage during long-term therapy was analysed excluding the data on AboBoNT (Dysport®). Therefore, data of 633 treatments in 52 patients was available. During initial therapy, significantly less dose of BoNT (initial dosage, mean 24.2 ± 10.2 units) was injected compared with the long-term treatments (average total dosage32.2 ± 13.4 units, $P < .0001$). Analysis of the BoNT dosage per visit revealed a continuous increase of the amount of injected toxin up to 19th-20th treatment (Figure 2; Friedman test, $P < .0001$). Men were treated with a significantly higher dose of BoNT (38 ± 13 units) than women (30 ± 13 units; $P < .0001$). Patients with synkinesis after an idiopathic palsy were injected with a significantly lower dose (28.3 ± 15 units) compared with the post-inflammatory (32.5 ± 12 units), post-traumatic (33.6 ± 6.2 units) or to an iatrogenic palsy (33.2 ± 12.3 units; $P < .0001$).

No significant changes in injected dosage per muscle during initial treatment compared with the long-term treatment were evident. Distribution of the injected BoNT dosage per muscle on the right and left side (initial treatment compared with the total number of treatments) is shown in the Tables S1 and S2.

3.5 | Time intervals between treatments

The median time interval between the visits was 2.9 months (95% CI 3.3-3.7) for IncoBoNT (95% CI 3.3-3.7) and OnaBoNT (95% CI 3.3-3.7) and 3.9 months for AboBoNT (95% CI 2.7-6 months). There was no significant difference in time period between injections with IncoBoNT, OnaBoNT or AboBoNT ($P = .153$).

TABLE 4 Distribution of treated muscles on the right and the left side in *patients with left-sided synkinesis* during the initial treatment (initial, n = 30) and in the long-term course (overall, n = 320)

	Right side			Left side		
	Initial	Overall	*P*-value	Initial	Overall	*P*-value
FRONTALIS	6.7%	25.0%	**.014**	13.3%	21.9%	.225
CORR	20.0%	29.4%	.224	33.3%	35.0%	.810
PROCERUS	33.3%	34.4%	.869	36.7%	37.2%	.917
ORB OCULI	3.3%	19.1%	**.022**	96.7%	96.6%	.953
NASALIS	0.0%	0.0%	1.000	6.7%	22.5%	**.028**
ZYGOM MAJ	0.0%	0.0%	1.000	33.3%	35.0%	.869
ORB ORIS	3.3%	5.9%	.519	3.3%	17.8%	**.028**
DEPRESS ANG ORIS	0.0%	5.3%	.171	3.3%	6.3%	.482
LEV LABII SUP	0.0%	0.0%	1.000	3.3%	1.6%	.418
MENTALIS	23.3%	40.9%	**.036**	63.3%	67.2%	.581
BUCCINATOR	0.0%	0.0%	1.000	0.0%	6.3%	.136
PLATYSMA	0.0%	2.2%	.388	46.7%	49.1%	.770
MASSETER	0.0%	0.0%	1.000	3.3%	1.6%	.418

Note: P-values were calculated with Mann-Whitney *U* test. Significant changes are highlighted (bold).

TABLE 5 Total dosage of BoNT during the initial treatment and in the long-term course in units

	N		Mean ± SD		Min.		Max.		*P*-value
	Initial	Total	Initial	Total	Initial	Total	Initial	Total	Mann-Whitney-*U*
OnaBoNT (Botox®)	24	282	26 ± 10.5	30.2 ± 13.6	7.5	1	50	65	.059
IncoBoNT (Xeomin®)	28	351	23.1 ± 9.9	33.8 ± 13.1	8	5	50	75	<.0001
AboBoNT (Dysport®)	11	50	55.4 ± 23.8	67.6 ± 45.1	20	20	90	240	.697

4 | DISCUSSION

4.1 | Synopsis of the key findings of the study

In this study population, dosage of IncoBoNT and OnaBoNT increased significantly during repeated treatments. The rising dosage correlated to the increasing number of injected muscles in the long-term course of therapy. At the same time, the BoNT dose that was injected per muscle remained stable. The BoNT dose nearly doubled after the first 20 consecutive treatments. Men were treated with a significantly higher BoNT dose than women. Furthermore, aetiology of facial nerve palsy significantly influenced the dosage. Patients with an idiopathic palsy needed significantly less toxin for synkinesis treatment. All these facts indicate that finding the right dosage in each patient should be individually customized and might take longer time than expected. Furthermore, the severity of synkinesis is very variable in each patient. Therefore, optimal therapy requires an individual adjustment of the injection pattern and an adjustment of the dose over time.

As defective healing often affects more than one muscle and is associated with more than just muscular symptoms, a complex therapy regime is required. In the present study, regardless of the side of paresis, a bilateral treatment was favoured in the long-term course of therapy. The synkinesis side was preferably injected during the initial visit. Our results confirm, that more complex injection patterns with higher number of mimic muscles on both sides is needed to reach a good result for the patient. Number of injected muscles increased significantly during the follow-up therapy (Figure 1). Increasing number of muscles may be a result of changing practice of the specialist. Experienced therapist may choose more "aggressive" approach for the first treatment. From 1998 to 2018 various practitioners were involved in patients' treatment in our department. For this reason, changed behaviour of a single investigator cannot be the only reason for the evolution of the injection pattern. The more probable reason for this change is the individual adjustment of pattern and dose over time, only reached after several visits.

Stable median time intervals of 12 weeks between the treatments indicate a stable duration of action of all three BoNT formulations in facial synkinesis. However, this evaluation was based on retrospective data. Several patients, especially the experienced ones, made fixed appointments for their next treatment. However, the treatment intervals indicate that the duration of action of BoNT in facial synkinesis is shorter than for cosmetic treatment (4 to 6months).[8-10] For example, the median duration of action of 12.4 weeks for OnaBoNT

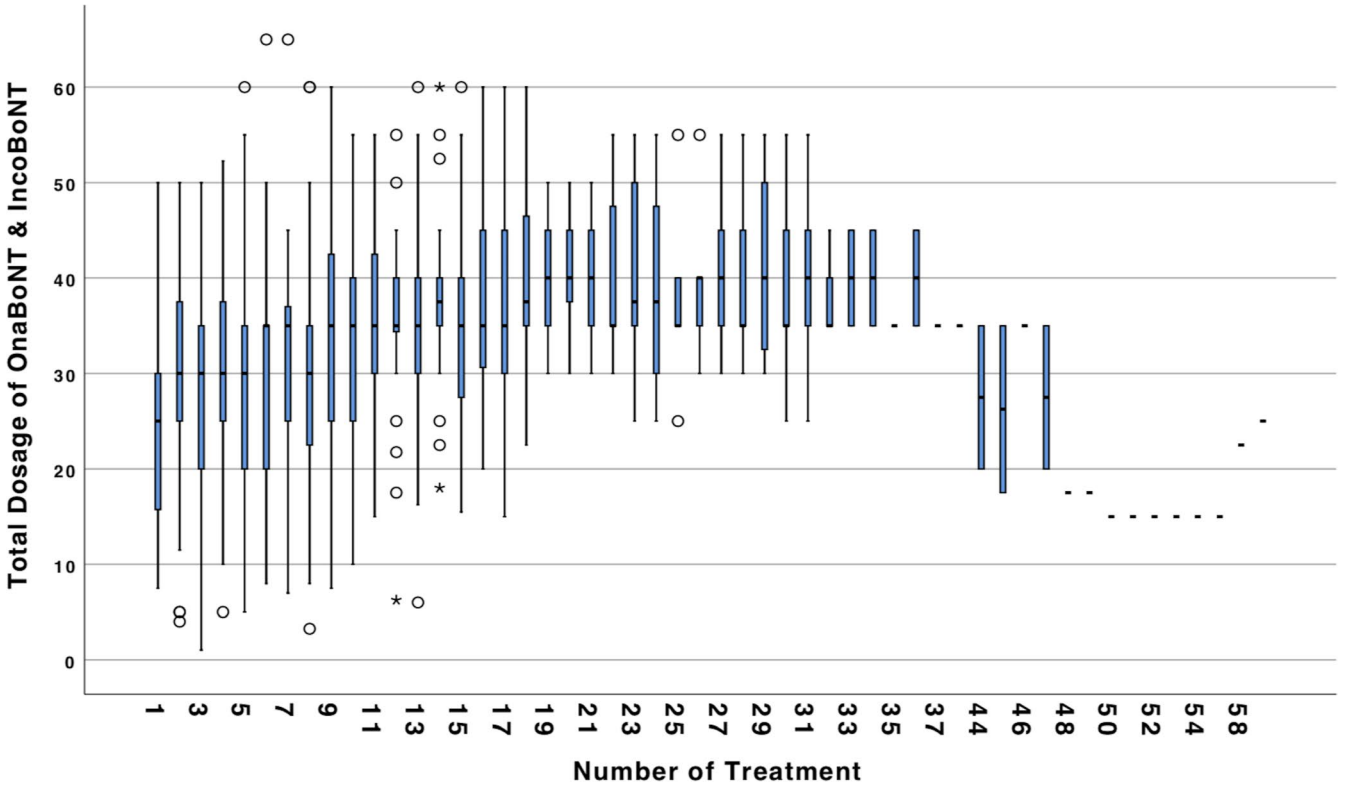


FIGURE 2 Distribution of dosage of OnaBoNT and IncoBoNT during the entire course of therapy. Continuous increase in dosage is evident during the first 19-20 treatments. Friedman test, $P < .0001$. (Group size for treatment number 35, 37,38, 46, 48-58 is N = 1. Group size for treatment number 36, 44, 45 and 47 is N = 2.)

corresponds rather to the subjectively reported median duration of response in patients with spasmodic dysphonia (11 weeks)[14] or with cervical dystonia (10 weeks)[15] and myofascial neck pain (10.4 weeks),[15] or to the recommended treatment interval for the chronic migraine (12 weeks).[16,17] Thus, results of this study strongly indicate that minimizing spasms and hypertonicity, synkinetic involuntary movements and pain relief via nociceptive effect of BoNT are some (but not all of them) leading targets of the therapy.

4.2 | Strength of the study

In the current study, we evaluated the long-term effect of therapy with all three available BoNT preparations in patients with facial synkinesis. Due to the long follow-up time of 20 years and large number of patients with high number of repeated visits we could analyse the change in injection pattern and in dosage of BoNT.

4.3 | Comparison with other studies

Several groups described an increase of BoNT dosage in patients with synkinetic defective healing and support our results.[5,12,18,19] Kollewe et al demonstrated an increase of BoNT dosage in 14% of the patients.[20] Laskawi et al described the significant increase of BoNT dosage in patients with synkinesis, blepharospasms and hemifacial spasms. He observed that in patients with synkinesis the increase of dose was associated with an increasing number of treated muscles. This was contrary to blepharospasm patients, in which the increasing dose correlated to the increasing dosage per injection point. In the present study, increasing dosage corresponded to higher number of treated muscles, whereas the amount of toxin per muscle remained the same.

Mean total dosage of OnaBoNT and IncoBoNT in our patients mainly correlated with the average amount of toxin applied elsewhere (30-34 units). Nevertheless, Choi et al emphasised use of a lower dosages of OnaBoNT. There are several reports showing that higher amounts of OnaBoNT are well tolerated and are associated with higher satisfaction scores and more favourable outcome.[12,20] Amount of AboBoNT (68 units) was higher in our patients than reported elsewhere.[20] It is difficult to compare results because of not standardised conversion dosage of ABoNT.[9,20,21] For this reason, we recommend careful analysis of the problem in each patient and an accurate adaptation of the injection pattern for each BoNT preparation to achieve the optimal result.

In our patients, injection of both, synkinetic and non-synkinetic side was favoured during long-term therapy. Worldwide, treatment of both sides is the favoured method of therapy.[1,12,20] However, the debate about the right side of treatment is still ongoing.[5,19] Synkinetic side was preferably treated during the initial visit in our centre. The reason for the change of injection pattern might be patients´ rising satisfaction with the result and diminishing fear of the toxin itself. After several effective treatments, patients might

follow the recommendations of the specialist more willingly. Several publication on quality of life after BoNT therapy observed increased satisfaction with the result in the long-term course of therapy and an increasing long-term compliance.[12,13,20]

Most frequently injected muscles in our patients (orbicularis oculi, procerus, mentalis and platysma) corresponded to the "core muscles" which were previously identified by Shinn et al[12] They also reported, that in average three treatments are needed to achieve a satisfactory result. However, an increase of dosage up to two years until a certain steady state is reached was also evident in their investigation.[12] In the present study, there was a continuous increase of dosage of BoNT over 20 consecutive treatments (corresponds to approximately 5 years). Even longer time periods were reported by Salles (up to 21 treatments) and Laskawi (up to 25 treatments).[5,19] This data indicates that the dose finding and achieving a steady state might be challenging in individual cases.[2]

Duration of effect of different toxin brands was previously discussed in several publications.[5,12,20] In our study, no significant difference in time periods between treatments with different BoNT brands could be observed. Based on previously published pro- and retrospective data, the same efficacy of all three toxins can be assumed.[3,20] Characterizing the use of BoNT-A in an aesthetic medicine, Flynn T. speculated that the duration of effect might change during repeated treatments.[8] The current study showed no significant differences in time intervals between treatments during long-term therapy of facial synkinesis. Several colleagues also support this result and showed that there is no evidence for loss of effectiveness during long-term therapy.[5,12,20]

In the area of the exclusively aesthetic application of BoNT for the treatment of glabella wrinkles or crow's feet, studies have shown a duration of effect of three to five months, which could be extended to six months in long-term treatment.[8,10]Most studies on synkinetic reinnervation showed a shorter duration of effect of BoNT-A.[12,20] These results underline the secondary role of aesthetic symmetry in treatment of synkinesis.

4.4 | Limitations of the study

Results of this study are biased because of the retrospective nature of data. To analyse the exact effect of duration of BoNT in facial synkinesis, prospective data assessment is required. Most of data on AboBoNT was collected between 1998 and 12/2004. Especially in this group, change in treatment dose is biased because of missing data. Furthermore, lack of exact standardised conversion of the equivalent dose for AboBoNT and OnaBoNT/IncoBoNT does not allow accurate comparison of these toxin preparations.[9,21]

No conclusions can be drawn regarding the subjective impact of the long-term therapy with BoNT on quality of life. Due to the retrospective nature of the data, an ad hoc analysis was not possible, because no standardised assessment was previously used in our centre.

4.5 | Clinical applicability of the study

The presented results provide an orientation in dose finding and injection regimen of BoNT in patients with facial synkinesis. The study shows the need for individual assessment and titration of dosage over time to achieve a satisfactory result.

5 | ETHICAL CONSIDERATION

For the retrospective analysis of patient-related data, the protocol of this study was reviewed and approved by the Ethics Committee of the University of Cologne. Study was registered under the internal number 19-1681.

CONFLICT OF INTEREST

All authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon request.

ORCID

Carina Pick https://orcid.org/0000-0002-5514-2167
Maria Grosheva https://orcid.org/0000-0002-5985-468X

REFERENCES

1. Choi KH, Rho SH, Lee JM, Jeon JH, Park SY, Kim J. Botulinum toxin injection of both sides of the face to treat post-paralytic facial synkinesis. *J Plast Reconstr Aesthetic Surg*. 2013;66(8):1058-1063.
2. Laskawi R. The use of botulinum toxin in head and face medicine: an interdisciplinary field. *Head Face Med*. 2008;4:5.
3. Laskawi R, Damenz W, Roggenkamper P, Baetz A. Botulinum toxin treatment in patients with facial synkinesis. *Eur Arch Otorhinolaryngol*. 1994;:S195-S199. https://doi.org/10.1007/978-3-642-85090-5_70.
4. Mehta RP, Hadlock TA. Botulinum toxin and quality of life in patients with facial paralysis. *Arch Facial Plast Surg*. 2008;10(2):84-87.
5. Salles AG, Da CEF, Ferreira MC, Remigio AFDN, Moraes LB, Gemperli R. Epidemiological overview of synkinesis in 353 patients with longstanding facial paralysis under treatment with botulinum toxin for 11 years. *Plast Reconstr Surg*. 2015;136(6):1289-1298.
6. Ackermann KH. Botox®-Behandlungen – immer noch keine Kassenleistung? *Urol*. 2017;56(3):367-368.
7. Dressler D, Vogt S, Saberi FA, Use O, Therapy BT, Use O. Der Off-Label Use am Beispiel der Botulinumtoxin-Therapie. *Klinische Neurophysiol*. 2016;47:87-91.
8. Flynn TC. Botulinum toxin: Examining duration of effect in facial aesthetic applications. *Am J Clin Dermatol*. 2010;11(3):183-199.
9. Carruthers A, Kane MAC, Flynn TC, et al. The Convergence of Medicine and Neurotoxins: A Focus on Botulinum Toxin Type A and Its Application in Aesthetic Medicine—A Global, Evidence-Based Botulinum Toxin Consensus Education Initiative. *Dermatol Surg*. 2013;39(3pt2):493-509.
10. Carruthers J, Fournier N, Kerscher M, Ruiz-Avila J, Trindade De Almeida AR, Kaeuper G. The Convergence of Medicine and Neurotoxins: A Focus on Botulinum Toxin Type A and Its Application in Aesthetic

Medicine—A Global, Evidence-Based Botulinum Toxin Consensus Education Initiative. *Dermatol Surg*. 2013;39(3pt2):510-525.
11. Filipo R, Spahiu I, Covelli E. Botulinum Toxin in the Treatment of Facial Synkinesis and Hyperkinesis. *Laryngoscope*. 2012;122(2):266-270.
12. Shinn JR, Nwabueze NN, Du L, et al. Treatment Patterns and Outcomes in Botulinum Therapy for Patients with Facial Synkinesis. *JAMA Facial Plast Surg*. 2019;21(3):244-251
13. Luijmes RE, Pouwels S, Beurskens CHG, Kleiss IJ, Siemann I, Ingels KJAO. Quality of life before and after different treatment modalities in peripheral facial palsy: A systematic review. *Laryngoscope*. 2017;127(5):1044-1051.
14. Novakovic D, Waters HH, D'Elia JB, Blitzer A. Botulinum toxin treatment of adductor spasmodic dysphonia: Longitudinal functional outcomes. *Laryngoscope*. 2011;121(3):606-612.
15. Diep D, Ko J, Lan J, Koprowicz KT, Ko G. Benefits, safety, and adjunct modality prevalences of long-term botulinum toxin injections for cervical dystonia and myofascial neck pain: A retrospective cohort study. *J Pain Res*. 2020;13:1297-1304.
16. Evans RW, Blumenfeld A. Botulinum Toxin Injections for Headache. *Headache*. 2003;43(6):682-685.
17. Winner PK, Blumenfeld AM, Eross EJ, et al. Long-Term Safety and Tolerability of OnabotulinumtoxinA Treatment in Patients with Chronic Migraine: Results of the COMPEL Study. *Drug Saf*. 2019;42(8):1013-1024.
18. Filipo R, Spahiu I, Covelli E, Nicastri M, Bertoli GA. Botulinum toxin in the treatment of facial synkinesis and hyperkinesis. *Laryngoscope*. 2012;122(2):266-270.
19. Laskawi R, Niemczewska A, Schneider S, Winterhoff J, Beutner C, Rohrbach S. Changes in dose and injection pattern in the botulinum toxin long-term therapy of facial dyskinesis. *Laryngorhinootologie*. 2014;93(3):186-192.
20. Kollewe K, Mohammadi B, Dengler R, Dressler D. Hemifacial spasm and reinnervation synkinesias: Long-term treatment with either Botox ® or Dysport ®. *J Neural Transm*. 2010;117(6):759-763.
21. Remigio AFDN, Salles AG, De Faria JCM, Ferreira MC. Comparison of the efficacy of onabotulinumtoxinA and abobotulinumtoxinA at the 1:3 conversion ratio for the treatment of asymmetry after long-term facial paralysis. *Plast Reconstr Surg*. 2015;135(1):239-249.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Alipour S, Pick C, Jansen S, Rink S, Klußmann JP, Grosheva M. Long-term therapy with botulinum toxin in facial synkinesis: Retrospective data analysis of data from 1998 to 2018. *Clin Otolaryngol*. 2021;00:1–9. https://doi.org/10.1111/coa.13729